

ST PAUL DOWNTOWN, MN, USA

WMO: 726584

Lat: 44.932N

Lon: 93.056W

Elev: 213

StdP: 98.79

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14927

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -23.2	(c) -21.0	(d) -27.9	(e) 0.3	(f) -22.4	(g) -26.0	(h) 0.4	(i) -20.5	(j) 11.7	(k) -9.9	(l) 10.6	(m) -10.0	(n) 3.6	(o) 310	(p) 0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	32.4	23.2	30.9	22.5	29.0	21.4	25.1	30.2	23.9	28.7	22.8	27.5	5.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.4	18.7	28.5	22.4	17.5	27.3	21.3	16.3	26.0	77.6	30.2	72.8	29.1	68.5	27.6	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.5	(b) 9.0	(c) 8.2	(e) -26.7	(f) 35.4	(g) 3.4	(h) 1.7	(i) -29.1	(j) 36.7	(k) -31.1	(l) 37.7	(m) -33.0	(n) 38.7	(o) -35.4	(p) 39.9		
(a) 10.5	(b) 9.0	(c) 8.2	(e) -27.0	(f) 27.0	(g) 3.3	(h) 1.3	(i) -29.3	(j) 27.9	(k) -31.2	(l) 28.7	(m) -33.1	(n) 29.4	(o) -35.4	(p) 30.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	-8.8	-6.4	0.6	8.4	15.1	20.5	23.2	21.8	17.6	9.9	1.8	-5.6
	DBStd	12.34	6.84	6.72	6.63	5.52	4.68	3.59	3.05	2.95	4.55	4.99	5.76	6.33
	HDD10.0	2242	582	458	299	93	9	0	0	0	3	63	252	482
	HDD18.3	4101	840	692	550	300	122	19	3	5	67	266	496	741
	CDD10.0	1611	0	0	8	46	168	316	411	367	231	60	6	0
	CDD18.3	427	0	0	1	2	22	86	154	114	45	4	0	0
	CDH23.3	3515	0	0	3	27	224	703	1351	836	336	36	0	0
	CDH26.7	1093	0	0	0	7	68	217	465	237	93	6	0	0
Wind	WSAvg	3.7	3.9	4.0	4.0	4.2	3.9	3.4	3.3	3.1	3.5	3.8	3.9	3.7
Precipitation	PrecAvg	841	21	18	44	75	106	125	114	116	84	69	44	27
	PrecMax	1038	59	39	103	160	221	236	284	222	169	143	133	74
	PrecMin	607	3	3	9	23	22	55	49	29	16	18	2	6
	PrecStd	111	13	9	23	33	48	48	54	52	49	39	35	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.2	11.3	21.4	27.2	32.2	33.9	35.4	33.5	32.1	27.4	19.1	10.1
		MCWB	3.5	7.6	14.4	16.5	20.4	23.6	25.2	23.6	22.2	18.8	12.9	7.3
	2%	DB	3.7	7.4	16.8	22.6	28.3	31.3	32.5	31.1	29.0	23.0	15.5	6.8
		MCWB	1.7	4.9	11.8	13.4	18.4	22.2	24.1	23.0	21.4	16.3	10.8	4.3
	5%	DB	2.2	4.5	12.9	20.0	26.0	29.0	31.0	29.0	27.3	20.5	12.5	4.0
		MCWB	0.8	2.2	8.8	11.8	17.2	20.9	23.1	21.9	20.2	14.9	8.6	2.2
	10%	DB	0.7	2.5	9.7	17.4	23.0	27.4	29.1	27.6	24.9	17.6	9.9	2.4
		MCWB	-0.5	0.8	6.3	10.3	15.5	19.5	21.9	20.8	19.1	13.1	6.8	1.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.6	7.8	15.9	17.5	22.7	25.3	27.1	26.2	24.3	20.7	14.6	8.5
		MCDB	6.3	10.4	19.5	25.6	29.2	31.5	31.8	31.4	28.9	24.5	17.9	9.2
	2%	WB	1.9	4.8	12.3	14.7	19.9	23.7	25.6	24.4	22.6	17.7	11.6	4.3
		MCDB	3.2	7.2	16.1	20.6	26.0	29.4	30.7	28.9	27.3	21.3	14.9	6.3
	5%	WB	0.6	2.6	9.1	12.8	18.4	22.1	24.3	23.3	21.4	15.6	8.9	2.5
		MCDB	2.0	4.5	12.8	18.7	23.6	27.4	29.1	27.4	25.6	19.4	11.8	4.0
	10%	WB	-0.7	0.8	6.3	11.1	16.9	20.8	23.1	22.1	20.0	13.6	6.9	1.0
		MCDB	0.4	2.5	9.3	16.6	21.8	25.4	28.0	26.2	23.9	17.4	9.7	2.3
Mean Daily Temperature Range	5% DB	MDBR	8.0	8.5	9.1	10.4	10.6	10.0	9.9	9.7	10.1	9.2	7.9	7.1
		MCDBR	9.2	9.7	13.3	14.6	14.1	12.5	11.5	11.2	12.2	13.6	11.7	9.3
		MCWBR	7.4	7.5	8.6	8.0	6.6	5.7	5.1	5.1	5.9	7.6	7.6	7.5
	5% WB	MCDBR	8.6	9.2	12.7	13.1	12.2	10.9	10.5	9.9	10.6	11.3	10.8	8.6
		MCWBR	7.2	7.2	8.6	8.0	6.7	6.0	5.7	5.7	6.1	7.4	7.6	7.1
Clear-Sky Solar Irradiance	taub	0.270	0.289	0.325	0.381	0.412	0.404	0.401	0.408	0.379	0.326	0.294	0.271	
	taud	2.415	2.343	2.368	2.284	2.251	2.319	2.348	2.332	2.377	2.518	2.511	2.435	
	Ebn at Noon	858	899	909	879	859	866	863	843	839	850	824	817	
	Edh at Noon	71	93	105	125	133	125	120	118	104	79	66	64	
All-Sky Solar Radiation	RadAvg	1.72	2.81	3.80	4.59	5.32	5.93	6.28	5.25	4.09	2.60	1.66	1.21	
	RadStd	0.20	0.29	0.35	0.47	0.43	0.47	0.23	0.37	0.29	0.35	0.15	0.13	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (57 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page